

Assessing the influence of text coherency and interval testing on the overall
retrieval process and transferred learning of a complex text

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Abstract

While the retrieval practice effect is well documented amongst the surrounding literature for enhancing long term retention, several studies argued seeing less robust effects for complex text materials (Karpicke & Aue; Hinze & Wiley, 2011). This study assessed how structural elements like text coherency and interval tests affect the depth and levels of transfer-appropriate processing required to skilfully recall knowledge from complex texts and apply learning into a novel MCQ final assessment. Results revealed that FITB tests allow more adept transfer of learning and recall when confronted with less coherent materials, whilst free recall only accomplished better transfer of learning and recall when confronted with far more coherent materials. This suggests that the organizational text structures presented in lectures or other similar mediums, and the testing techniques employed for studying can massively influence students' academic abilities.

Recently, a vast amount of literature has emerged, highlighting the potential benefits of employing retrieval practices or *interval tests* on boosting long-term retention and memory encoding abilities during learning (Roediger & Karpicke 2006, cited by Hinze and Wiley, 2011; de Jonge *et al.*, 2015). Retrieval practices are active test strategies that help solidify our memory consolidation processes. The frequent application of retrieval practices over prolonged intervals tends to attenuate rates of forgetfulness by strengthening “elaborate memory connections and traces” of learned content from materials (Carpenter *et al.*, 2009, cited by de Jonge *et al.*, 2015). Hence, many studies have insisted that retrieval practices pose a higher advantage over just passively restudying when trying to excel in educational settings (Zaromb and Roediger, 2010; Hinze & Wiley, 2011).

However, the retrieval practice effect appeared to be slightly erratic in some studies which tested more content-heavy texts such as ones from simulated lectures, prose, and textbooks readings (de Jonge *et al.*, 2015; Tran *et al.*, 2015, cited by Karpicke & Aue, 2015). This shortcoming could be attributed to the retrieval practices used. For example, de Jonge (2015)’s study implemented filling-in-the-blank (FITB) practices to compare the relative test benefits obtained from incoherent and coherent text versions. Unintentionally, this might have “prompted superficial, item-specific processing” where instead, participants prioritized more time revisiting specific words or phrases they thought would more likely be blanked out (Hinze & Wiley, 2011; Vauras *et al.*, 1992). Whilst other studies like Hinze & Wiley (2011)’s yielded more robust and encouraging results through experimenting with more open-ended test formats such as paragraph recall (Karpicke & Aue, 2015). Open-ended test formats empower participants to produce more elaborated responses which can further enhance one’s ability to encode deeply and prompt quicker and far accurate retrieval of learned material content for future transfer of learning (Hinze & Wiley, 2011). Indeed, the derived benefit from retrieval practices does heavily depend on the depth and transfer appropriate processing acquired through testing from different formats.

Applied learning or *transfer of learning* encompasses the critical ability required to generalize previously learning into new unexplored contexts and hinges on the ability to skilfully recall knowledge even when there is not much understanding of the full picture sometimes. Although de Jonge (2015)'s results detailed the impacts of testing retrieval and transferred learning on different text coherencies from FITB, a key element it failed to focus on was which retrieval practice best magnified the relative benefits of testing and applied learning. Yet while Hinze and Wiley (2011) successfully captured that limitation in their last experiment, they foregone the opportunity to test its respective impacts on incoherent text materials. By intertwining the pivotal elements of both studies, this paper will assess:

How does text coherency and interval tests impact the overall memory retrieval processes and transfer learning of complex materials?

Incoherent materials remain complex to read, with little connectivity or “referential ties” between consecutive sentences (Vauras et al., 1992). Additionally, more processing time is required to understand semantic and structural schema underlying the text. Therefore, it is predicted that those taking FITB as an interval test will do significantly better than those either attempting free recall (paragraph) or just passively restudying the incoherent text before the final assessment.

Whilst conventional and logically organized coherent materials allow for more effective interpretation, construction, and active recall (Vauras et al., 1992). Thus, it is hypothesized that those doing free recall (paragraph) as an interval test will gain ground over those restudying or have completed FITB before the final assessment for the coherent text.

Results

As seen from Figure 1, participants who restudied the incoherent text material were significantly worse at transferring learned knowledge into the final MCQ than participants who were tested by FITB, $p = .03$.

Yet, when compared to participants doing a free recall, the participants who restudied the incoherent text material did significantly better in the final MCQ, $p = .04$.

Conversely, for the coherent text material, participants who attempted free recall did significantly better than participants who restudied before being tested in the final MCQ, $p = .01$.

Moreover, participants who attempted free recall did significantly better in the final test than participants who were tested by FITB on the coherent text material, $p = .01$, as seen from Figure 1. No other differences were significant (all other p 's $> .05$).

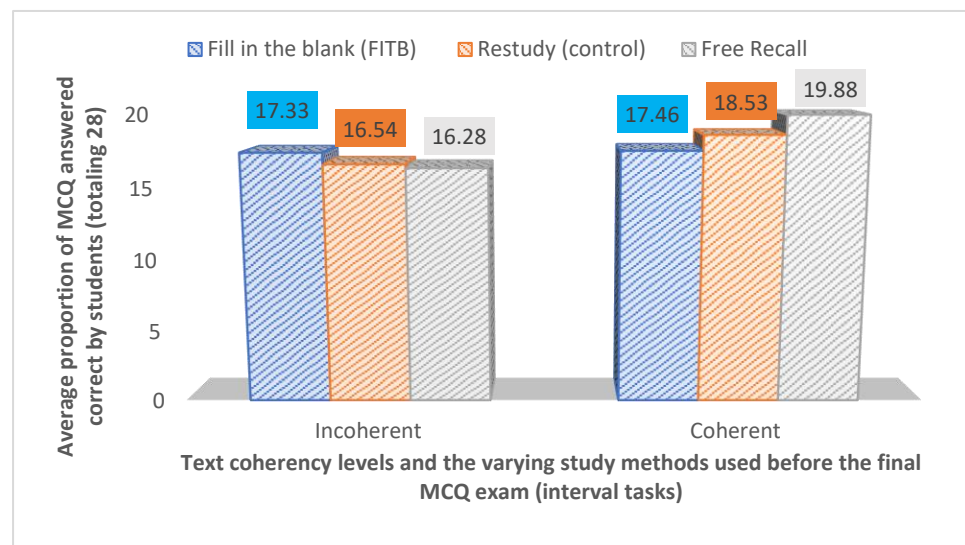


Figure 1. Final test performance on MCQ test after completing interval tasks relating to the incoherent and coherent text versions of the assessed material.

Discussion

As anticipated, the results corroborated with our hypotheses. Unexpectedly, both findings fit well with the transfer-appropriate processing theory.

For the incoherent text, FITB is better adapted at transferring learning and recalling information than free recall since the FITB format practically encourages participants to understand the assigned text in a shallow way by selectively skimming through to retrieve the correct answer (Zaromb & Roediger, 2010; Karpicke & Aue, 2015).

Whilst free recall optimised transfer learning and the overall retrieval effect better than FITB for coherently structured texts since it predominantly relies on deeply processing meaningful links between sentences to retrieve information stronger which is quite daunting to accomplish when studying incoherent texts since consecutive sentences are deliberately jumbled to reduce 'referential ties' (Hinze & Wiley, 2011; de Jonge et al., 2015; Vauras et al., 1992).

Although our study improved vastly by incorporating another test strategy for a final assessment to reduce bias, the final MCQ may have impaired results as through recognition better results can be scored. Thus, a better recommendation might be to instead replace it with a short answer question test.

Furthermore, future studies could investigate:

- Effect of selective attention spans, motivation, domain knowledge and language proficiency on retrieval abilities
- Replicating educational settings of having final exams further down the road by assessing their respective effects within spaced, longer intervals
- Other commonly used tactics (i.e chunking, mnemonics) employed to improve retrieval of less coherent texts

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